Predictive Analytics Poster Rubric (Poster-Only Assessment)

#	Criterion	Points	Exemplary (Full Points)	Competent (Partial Points)	Needs Improvement (Minimal Points)
1	Prediction Goal(s)	5	5 pts: Clear, specific predictive objective(s); response type (regression/classification) explicitly identified and justified; names the stakeholder and the decision the predictions support; strong business/research rationale.	3–4 pts: Goal and response type stated, but the stakeholder, the decision, or the rationale is thin; limited link to context.	0–2 pts: Goal unclear/missing; response type unspecified; no link to a stakeholder's decision.
2	Dataset Overview	8	7–8 pts: Reports #obs/#features; types for key features; response variable identified/typed; data provenance/window stated.	4–6 pts: Size and response identified; partial feature typing; minimal provenance.	0–3 pts: Missing size/response ID; little/no feature typing or context.
3	Exploratory Data Analysis (EDA)	12	10–12 pts: Appropriate summaries (response + key predictors); ≥2 relevant plots; interprets skew/outliers/imbalance/seasonality; connects insights to modeling choices and metric selection.	6–9 pts: Basic stats and ≥1 plot; some interpretation but limited linkage to modeling.	0–5 pts: Incomplete stats; weak/irrelevant visuals; minimal/no interpretation.
4	Data Preparation (Feature Engineering, Missing Data, etc.)	18	15–18 pts: ≥1 non-trivial new feature (interactions, lags/rolling stats, domain-driven ratios, text-derived signals) with clear rationale; quantifies missingness by feature; justified strategy (drop/impute/advanced) with leakage-safe implementation; demonstrates impact via ablation/CV deltas or sensitivity to imputation choices.	9–14 pts: New feature with partial rationale; identifies missingness and applies a reasonable method; limited evidence of impact or sensitivity analysis.	0–8 pts: No new/trivial feature; missingness unreported or mishandled; unjustified/inappropriate method; no evidence of value.
5	Baseline Model (Choice, CV, Interpretation)	17	14–17 pts: Appropriate simple baseline (e.g., Linear/Logistic Regression), the reference any complex model must beat; pipeline is leakage-safe; k=5/10 CV (time-respecting for forecasting) with a primary metric chosen for the cost of errors; reports the CV mean with its SD and approximate 95% interval, labeled descriptive (notebook 08); summarizes implications and limitations; proposes concrete next steps.	8–13 pts: Reasonable model; CV present with partial integration/details; metric adequate; interpretation limited or generic.	0–7 pts: Inappropriate/unspecified baseline; no/incorrect CV; unclear metric; raw numbers only; no reflection.

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6	Model Comparison and Champion Selection (Choice, Tuning, Paired Comparison, Rationale)	25	22–25 pts: Model aligned to data/goal (e.g., RF/GBM/SVM/NN); tuning inside CV with leakage controls. Before comparing: the primary metric and a practical margin (the smallest gain that would change the stakeholder's decision, with its business meaning). Baseline and tuned model (and any other finalist) compared on identical folds (same splitter and seed): per-fold paired differences and their mean, with the verdict named correctly under notebook 08's rule: real difference (every fold favors the same model and the mean gap clears the margin), no winner (the sign changes across folds), or practically equivalent (same sign but under the margin, so the simpler model wins). One champion, chosen as notebook 14 teaches (the baseline can win); extra complexity only on a real difference; rationale in the stakeholder's terms (what the gain means, secondary checks, runner-up). Interval overlap does not pick the champion; the test set is not used to select.	13–21 pts: Plausible model and tuning, but the comparison or decision is incomplete or misapplied: folds not paired; margin missing, without business meaning, or set after the results without saying so; verdict from means alone or from whether two intervals overlap; fold signs that disagree read as a win; or a champion without a rationale tied to the decision.	0–12 pts: Misaligned/unspecified model; superficial/absent tuning; incorrect CV; no clear comparison and no justified champion.
7	Report Quality & Visual Communication	10	9–10 pts: Clear, scannable poster; logical flow (Intro → Methods → Results → Conclusions); the champion and its main reason are easy to find; legible, labeled figures/tables (titles, axes, units, legends); consistent typography/layout; code refs/appendix where appropriate; minimal grammar issues.	6–8 pts: Generally readable; minor layout/labeling gaps; some figures under-explained or styling inconsistencies.	0–5 pts: Disorganized; illegible/unlabeled visuals; weak linkage between text and graphics; notable grammar/formatting issues.
8	Sources, Acknowledgments, AI Use & Data Privacy	5	5 pts: Data source cited (provider, URL or DOI, version or access date); references support the business-problem claims and the main software is credited (e.g., scikit-learn, pandas); Acknowledgments section names the Mitch Daniels School of Business, Prof. Moreira, the TAs, and other contributors; AI Use Disclosure names each AI tool and what it was used for and states that the team verified the output (or states that no AI tools were used); only aggregated or anonymized data and examples.	3–4 pts: AI Use Disclosure present but vague (names tools without saying what they did), or one other element missing or incomplete (e.g., acknowledgments omit the TAs; data source cited without references or tools). No privacy issue.	0–2 pts: No AI Use Disclosure, or two or more other elements missing. Any personally identifiable or NDA-restricted data on the poster scores 0 here and must be removed before printing.

Total: 100 points. This rubric grades M09 (poster first draft), M10 (final poster), and the Instructor/TA evaluation after the conference.

Criterion 6 at M09 (due Nov 1): the comparison and the champion get feedback, not a deduction, on the first draft; they are scored from M10.

Criterion 6 follows the notebooks: the paired comparison and margin from notebook 08 and the champion selection from notebook 14. If your margin was set after the M06/M08 results, say so on the poster: honest disclosure is not penalized. A baseline champion or an honest “no winner” verdict can earn full credit. No new model search or test-set opening is required.

AI use: per the syllabus AI Policy, every poster carries an AI Use Disclosure (the template has a section for it). Errors in AI-assisted work are the team’s responsibility.

Honors: a labeled honors section, where present, is graded only under the Honors Research Extension and is not scored under this rubric.